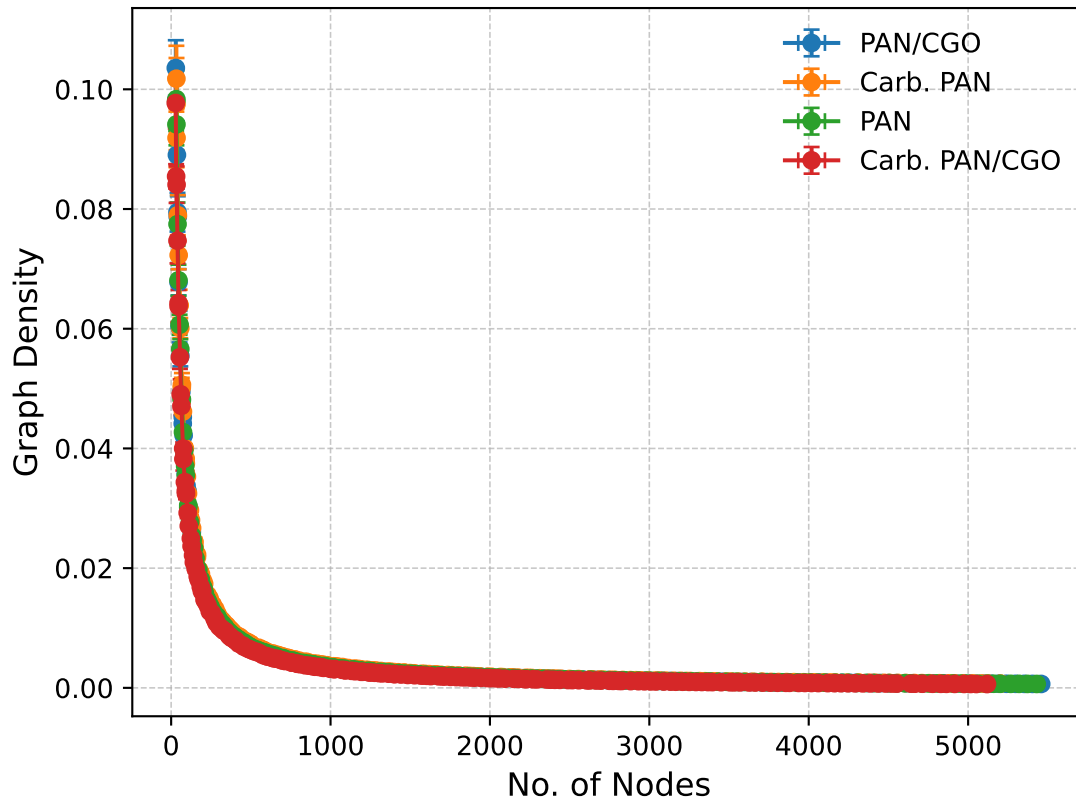


Nodes vs Graph Density (Actual Data)



Kolmogorov-Smirnov & P-Values

PAN/CGO:

Weibulll → KS=42.024, p=0.011
 Inverse Gaussian → KS=16.876, p=0.000
 Generalized Pareto → KS=1.215, p=0.051

Carb. PAN:

Weibulll → KS=9.958, p=0.786
 Inverse Gaussian → KS=16.337, p=0.000
 Generalized Pareto → KS=0.391, p=0.707

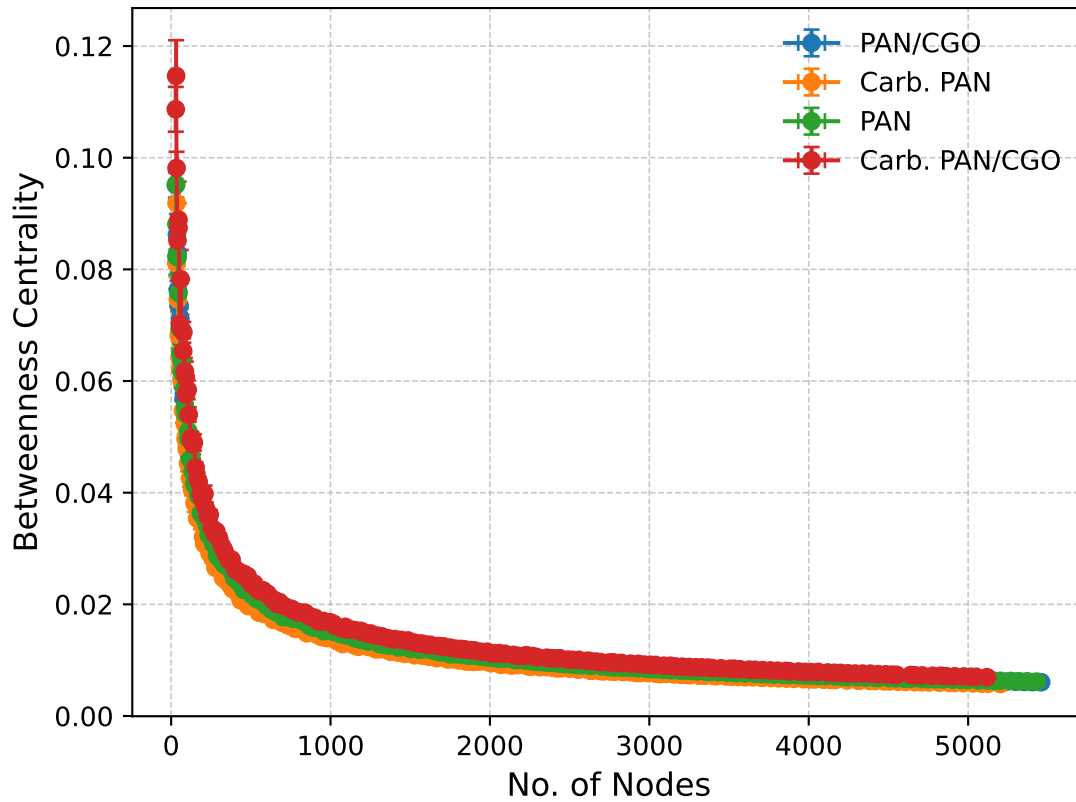
PAN:

Weibulll → KS=30.124, p=0.085
 Inverse Gaussian → KS=17.187, p=0.000
 Generalized Pareto → KS=1.255, p=0.047

Carb. PAN/CGO:

Weibulll → KS=43.818, p=0.013
 Inverse Gaussian → KS=17.206, p=0.000
 Generalized Pareto → KS=0.399, p=0.870

Nodes vs Betweenness Centrality (Actual Data)



Kolmogorov-Smirnov & P-Values

PAN/CGO:

Weibulll → KS=1.000, p=0.472
 Inverse Gaussian → KS=3.597, p=0.000
 Generalized Pareto → KS=0.978, p=0.105

Carb. PAN:

Weibulll → KS=1.307, p=0.366
 Inverse Gaussian → KS=3.694, p=0.000
 Generalized Pareto → KS=0.564, p=0.407

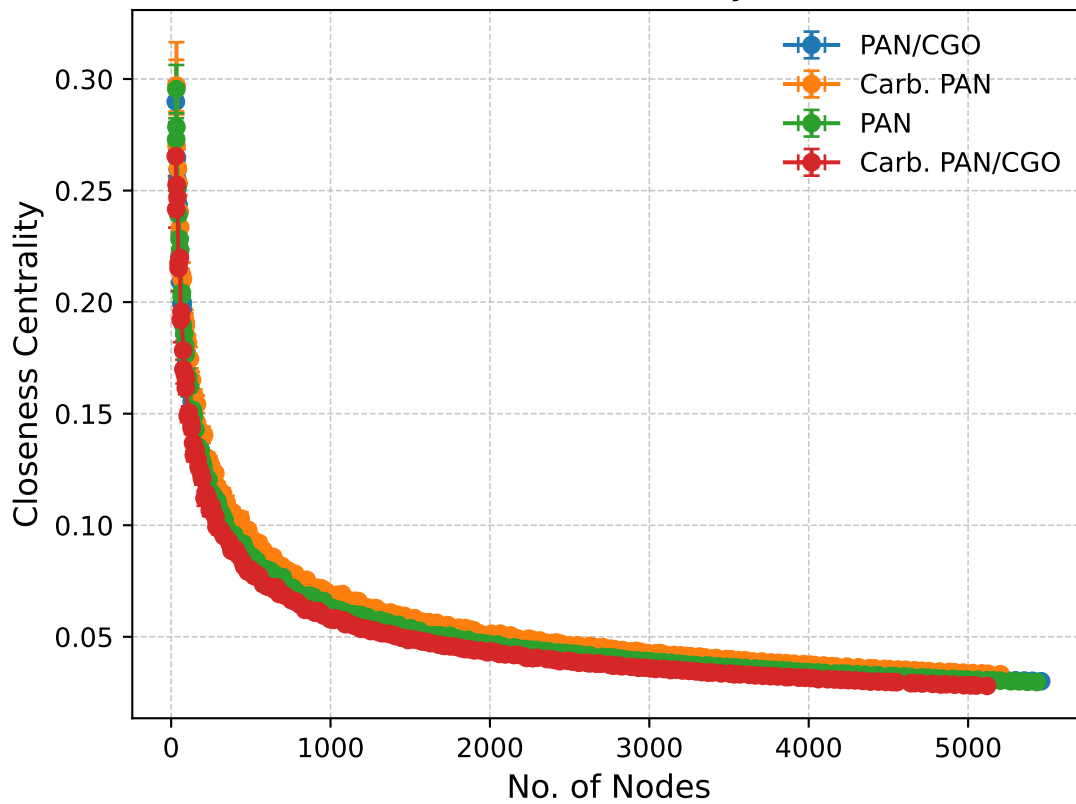
PAN:

Weibulll → KS=0.742, p=0.605
 Inverse Gaussian → KS=3.749, p=0.000
 Generalized Pareto → KS=1.228, p=0.048

Carb. PAN/CGO:

Weibulll → KS=0.655, p=0.685
 Inverse Gaussian → KS=3.892, p=0.000
 Generalized Pareto → KS=3.489, p=0.000

Nodes vs Closeness Centrality (Actual Data)



Kolmogorov-Smirnov & P-Values

PAN/CGO:

Weibull $\rightarrow$ KS=1.233, $p=0.297$
 Inverse Gaussian $\rightarrow$ KS=1.966, $p=0.001$
 Generalized Pareto $\rightarrow$ KS=0.507, $p=0.666$

Carb. PAN:

Weibull $\rightarrow$ KS=0.361, $p=0.967$
 Inverse Gaussian $\rightarrow$ KS=1.740, $p=0.002$
 Generalized Pareto $\rightarrow$ KS=0.497, $p=0.648$

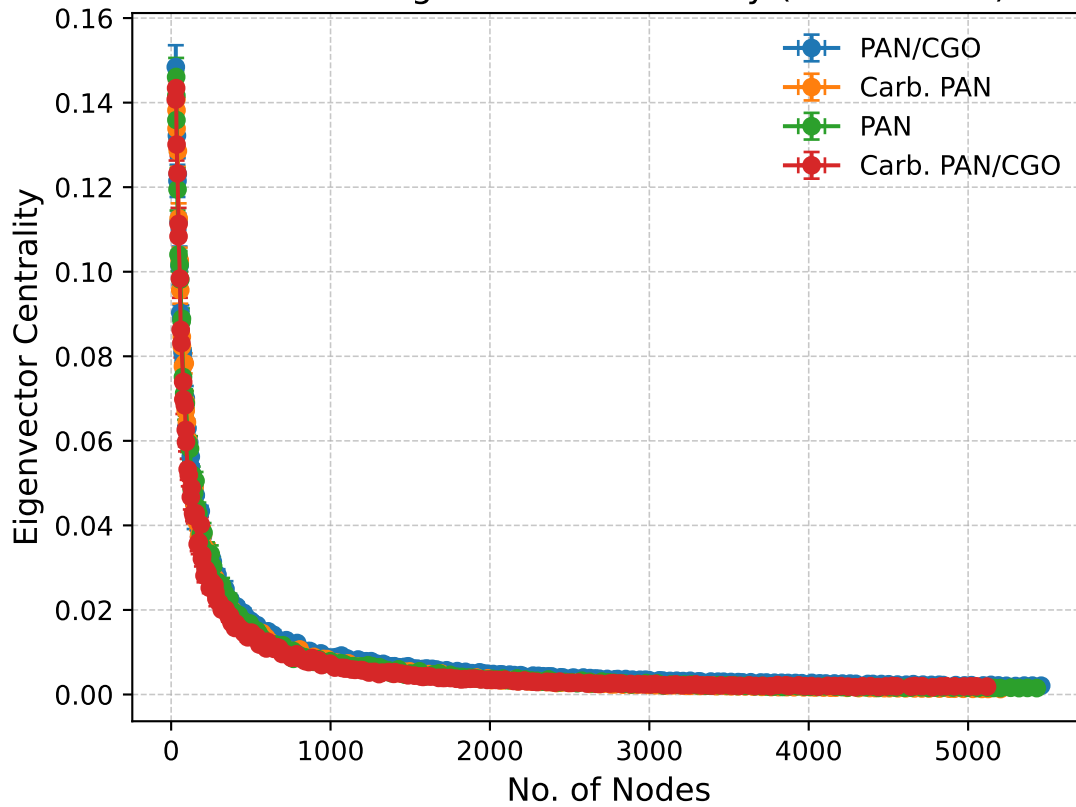
PAN:

Weibull $\rightarrow$ KS=1.723, $p=0.224$
 Inverse Gaussian $\rightarrow$ KS=2.003, $p=0.001$
 Generalized Pareto $\rightarrow$ KS=10.598, $p=0.000$

Carb. PAN/CGO:

Weibull $\rightarrow$ KS=2.540, $p=0.143$
 Inverse Gaussian $\rightarrow$ KS=2.033, $p=0.001$
 Generalized Pareto $\rightarrow$ KS=0.552, $p=0.625$

Nodes vs Eigenvector Centrality (Actual Data)



Kolmogorov-Smirnov & P-Values

PAN/CGO:

Weibull $\rightarrow$ KS=31.282, $p=0.041$
 Inverse Gaussian $\rightarrow$ KS=11.765, $p=0.000$
 Generalized Pareto $\rightarrow$ KS=0.389, $p=0.702$

Carb. PAN:

Weibull $\rightarrow$ KS=12.305, $p=0.559$
 Inverse Gaussian $\rightarrow$ KS=13.832, $p=0.000$
 Generalized Pareto $\rightarrow$ KS=0.719, $p=0.231$

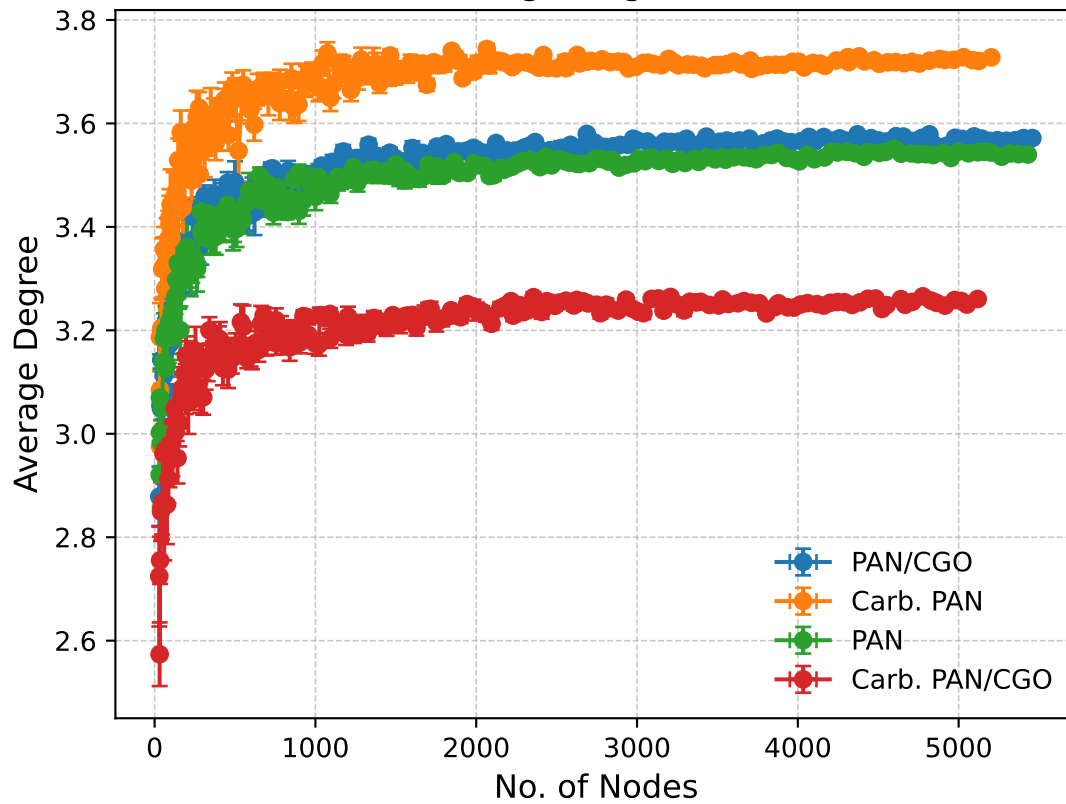
PAN:

Weibull $\rightarrow$ KS=12.128, $p=0.775$
 Inverse Gaussian $\rightarrow$ KS=15.255, $p=0.000$
 Generalized Pareto $\rightarrow$ KS=1.482, $p=0.021$

Carb. PAN/CGO:

Weibull $\rightarrow$ KS=26.158, $p=0.146$
 Inverse Gaussian $\rightarrow$ KS=17.389, $p=0.000$
 Generalized Pareto $\rightarrow$ KS=1.465, $p=0.023$

Nodes vs Average Degree (Actual Data)



Kolmogorov-Smirnov & P-Values

PAN/CGO:

Weibull → KS=10.292, p=0.006

Inverse Gaussian → KS=43.996, p=0.000

Generalized Pareto → KS=44.132, p=0.804

Carb. PAN:

Weibull → KS=44.940, p=0.000

Inverse Gaussian → KS=47.299, p=0.000

Generalized Pareto → KS=75.933, p=0.630

PAN:

Weibull → KS=11.203, p=0.004

Inverse Gaussian → KS=41.485, p=0.000

Generalized Pareto → KS=inf, p=0.000

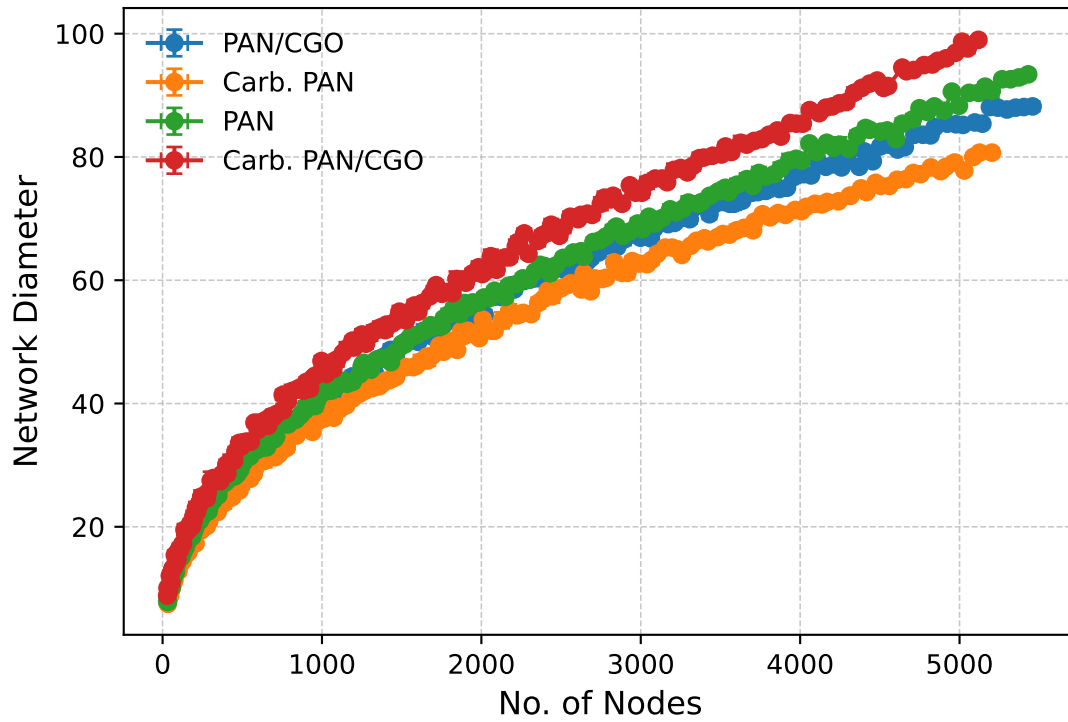
Carb. PAN/CGO:

Weibull → KS=10.213, p=0.008

Inverse Gaussian → KS=47.210, p=0.000

Generalized Pareto → KS=46.155, p=0.306

Nodes vs Network Diameter (Actual Data)

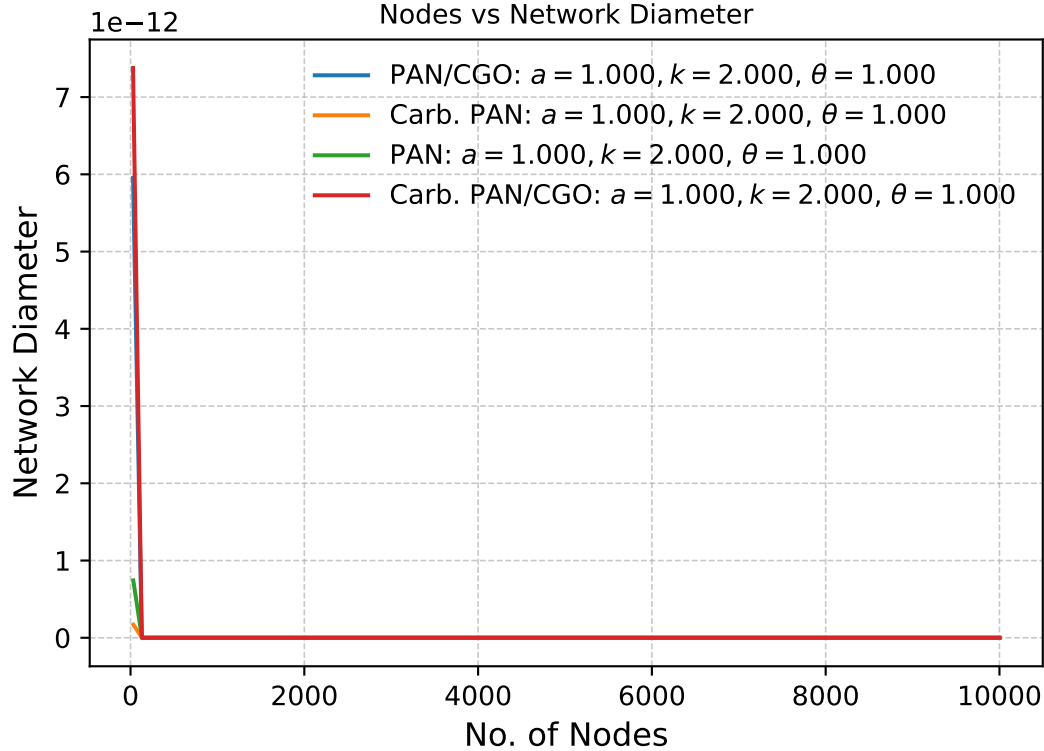


Kolmogorov-Smirnov & P-Values

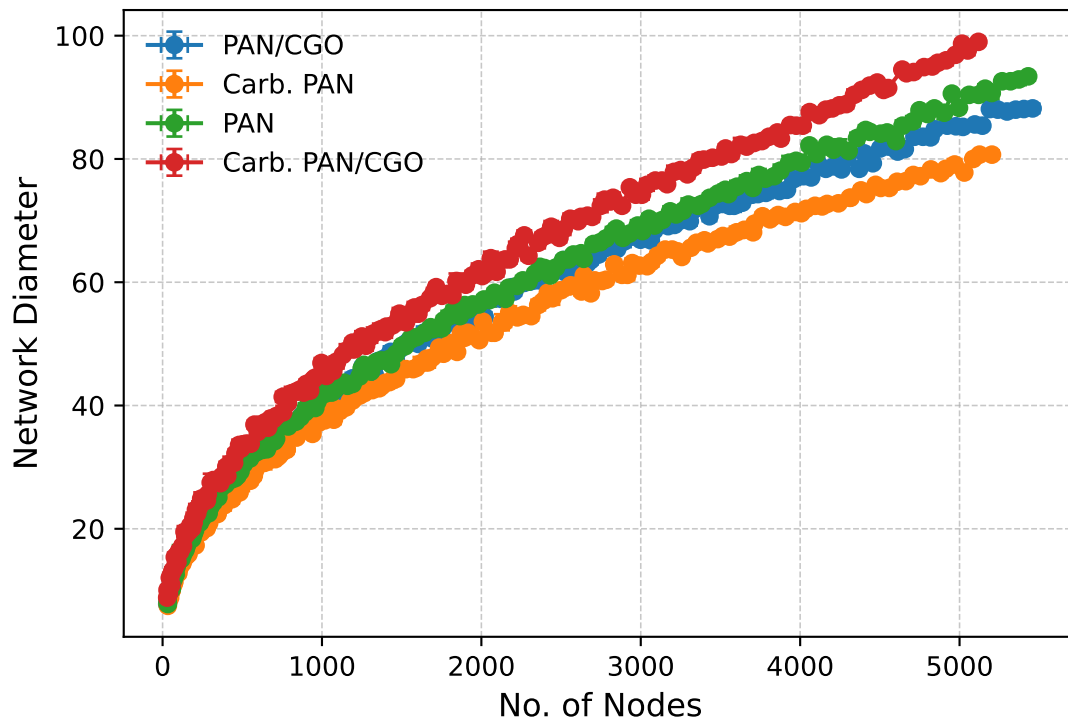
Goodness-of-fit tests skipped.

Gamma Fit: $y = a \cdot x^{-k} \cdot \exp\left(-\frac{x}{a}\right)$

Nodes vs Network Diameter



Nodes vs Network Diameter (Actual Data)



Kolmogorov-Smirnov & P-Values

PAN/CGO:
 Weibull $\rightarrow$ KS=2.077, p=0.067
 Inverse Gaussian $\rightarrow$ KS=8.395, p=0.000
 Generalized Pareto $\rightarrow$ KS=2.285, p=0.220

Carb. PAN:
 Weibull $\rightarrow$ KS=2.197, p=0.067
 Inverse Gaussian $\rightarrow$ KS=8.368, p=0.000
 Generalized Pareto $\rightarrow$ KS=2.817, p=0.114

PAN:
 Weibull $\rightarrow$ KS=1.825, p=0.084
 Inverse Gaussian $\rightarrow$ KS=7.965, p=0.000
 Generalized Pareto $\rightarrow$ KS=2.525, p=0.202

Carb. PAN/CGO:
 Weibull $\rightarrow$ KS=1.830, p=0.076
 Inverse Gaussian $\rightarrow$ KS=8.198, p=0.000
 Generalized Pareto $\rightarrow$ KS=2.891, p=0.543

Gamma Fit: $y = a \cdot x^{-k} \cdot \exp(-\frac{x}{a})$ Nodes vs Network Diameter

